

**King Fahd University of Petroleum and Minerals**  
**Department of Mathematics**

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**Math 208    Major Exam I    Term 253**

Original exam date: 2 July 2026.    Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

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**1.** Which one of the following subsets is a subspace of  $\mathbb{R}^4$ ?

- (a) The set of all vectors  $(x_1, x_2, x_3, x_4)$  such that  $x_1 \geq 0$
- (b) The set of all vectors  $(x_1, x_2, x_3, x_4)$  such that  $x_1^2 + x_2^2 = 1$
- (c) The set of all vectors  $(x_1, x_2, x_3, x_4)$  such that  $x_1 + x_2 - x_3 + 2x_4 = 1$
- (d) The set of all vectors  $(x_1, x_2, x_3, x_4)$  such that  $x_1 - 2x_2 = 0$  and  $x_3 + 5x_4 = 0$
- (e) The set of all vectors  $(x_1, x_2, x_3, x_4)$  such that  $x_1x_3 = 0$

**2.** If the function  $g(x)$  with  $g(0) = 1$  makes the differential equation  $[g(x) - x \cos(xy)] dy = [e^{2x} + y \cos(xy)] dx$  an exact differential equation, then  $g(x) =$

- (a)  $e^{2x}$
- (b)  $\sin x + 1$
- (c) 1
- (d)  $\cos(2x)$
- (e)  $\cos x$

**3.** The sum of all values of  $m$  for which  $y = x^m$  is a solution of the differential equation  $x^2y'' - 7xy' + 15y = 0$  is

- (a) 8
- (b) 0
- (c) -2
- (d) -8
- (e) 2

4. A thermometer with an initial reading of  $10^\circ$  is brought into a room kept at a constant  $30^\circ$ . If the thermometer reads  $20^\circ$  after 2 minutes, what is the total time required for the thermometer to reach  $25^\circ$ ?

- (a) 4 minutes
- (b) 6 minutes
- (c) 8 minutes
- (d) 5 minutes
- (e) 3 minutes

5. If the vectors  $\mathbf{u} = (1, k, 2)$ ,  $\mathbf{v} = (2, 2, 0)$  and  $\mathbf{w} = (1, 2, -2)$  in  $\mathbb{R}^3$  are linearly dependent, then  $k =$

- (a) 1
- (b)  $-1$
- (c) 0
- (d) 2
- (e)  $-2$

6. The general solution of the differential equation  $\frac{dy}{dx} + y \cot x = \sin x$ ,  $0 < x < \pi$  is

- (a)  $4y \sin x = x - 2 \sin(2x) + c$
- (b)  $4y \sin x = 2x - \sin(2x) + c$
- (c)  $2y \sin x = x - \sin(2x) + c$
- (d)  $2y \sin x = 2x + \sin(2x) + c$
- (e)  $3y \sin x = 4x - \sin(2x) + c$

7. The solution of the linear differential equation  $\frac{dy}{dx} = \frac{y}{2y \ln y + y - x}$  is given by

- (a)  $xy = cx + y^2 \ln y$
- (b)  $2xy = c + y^2 \ln y$
- (c)  $xy = c + x^2 \ln y$
- (d)  $xy = c + y^2 \ln y$
- (e)  $2xy = cx + y^2 \ln y$

8. The general solution of the differential equation  $(x^2 + 3xy + y^2) dx - x^2 dy = 0$  is given by

(a)  $\ln|x| + \left(1 + \frac{y}{x}\right)^{-1} = c$

(b)  $\ln|x| + \frac{y}{x} + 1 = c$

(c)  $y = 2x \ln|x| + cx$

(d)  $\ln|x| + \left(1 + \frac{x}{y}\right)^{-1} = c$

(e)  $y = x \ln|x| + cx$

9. A general solution of the exact differential equation  $\left(\frac{2x}{y} - \frac{3y^2}{x^4}\right) dx + \left(\frac{2y}{x^3} - \frac{x^2}{y^2} + \frac{1}{\sqrt{y}}\right) dy = 0$  is

(a)  $\frac{x^2}{y} - \frac{y^2}{x^2} + 2\sqrt{y} = c$

(b)  $\frac{x}{y} + \frac{y^2}{x^3} + 2\sqrt{y} = c$

(c)  $\frac{x^2}{y} + \frac{y^2}{x^3} + 2\sqrt{y} = c$

(d)  $\frac{x^2}{y} - \frac{y}{x^3} + \sqrt{y} = c$

(e)  $\frac{x^2}{y^2} + \frac{y^2}{x^3} + \sqrt{y} = c$

10. The general solution of the differential equation  $xy'' + y' = 4x$ ,  $x > 0$  is given by

(a)  $y(x) = 2x^2 + Ax \ln x + B$

(b)  $y(x) = 2x^2 + A \ln x + B$

(c)  $y(x) = 4x^2 + Ax \ln x + B$

(d)  $y(x) = x^2 + A \ln x + B$

(e)  $y(x) = 3x^2 + A \ln x + B$

11. If a particle is moving in a straight line with acceleration  $a(t) = \frac{1}{\sqrt{1-t^2}}$ , initial position  $x(0) = 1$ , and initial velocity  $v(0) = 0$ , then the position of the particle at any time  $t$  is given by

(a)  $x(t) = t \sin^{-1} t + 3\sqrt{1-t^2}$

(b)  $x(t) = 2t \sin^{-1} t + 4\sqrt{1-t^2}$

(c)  $x(t) = t \sin^{-1} t + \sqrt{1-t^2}$

(d)  $x(t) = t \sin^{-1} t + 2\sqrt{1-t^2}$

(e)  $x(t) = 2t \sin^{-1} t + \sqrt{1-t^2}$

12. If the solution space of the system

$$\begin{aligned}x_1 - 2x_2 + x_3 + x_4 &= 0 \\x_1 - 2x_2 + 2x_3 - x_4 &= 0\end{aligned}$$

is the set of all linear combinations of the form  $s\mathbf{u} + t\mathbf{v}$ , where  $s, t$  are real numbers and  $\mathbf{u} = (a, b, 0, 0)$ ,  $\mathbf{v} = (m, n, 2, l)$ , then  $\frac{a}{b} + \frac{2}{l} =$

- (a) 2
- (b) 1
- (c) 0
- (d) 4
- (e) 3

13. By using an appropriate substitution, the differential equation  $\frac{dy}{dx} = 1 + \sqrt{y - 4x + 1}$  transforms into the separable differential equation

- (a)  $\frac{du}{\sqrt{u} + 3} = dx$
- (b)  $\frac{du}{\sqrt{u} - 3} = dx$
- (c)  $\frac{du}{\sqrt{u} + 2} = 3 dx$
- (d)  $\frac{du}{\sqrt{u} + 1} = dx$
- (e)  $\frac{du}{\sqrt{u} - 1} = 2 dx$

14. The solution of the initial-value problem  $x \frac{dy}{dx} - y = y^2 \ln x$ ,  $y(1) = \frac{1}{2}$  is given by

- (a)  $\frac{1}{y} = 1 - 2 \ln x + \frac{1}{x}$
- (b)  $\frac{1}{y} = 1 + 2 \ln x + \frac{1}{x}$
- (c)  $\frac{1}{y} = 1 - \ln x + \frac{1}{x}$
- (d)  $y = 1 + 2 \ln x - \frac{1}{2x}$
- (e)  $y = 1 - \ln x + \frac{1}{x}$

**15.** If  $y = y(x)$  is the solution of the initial value problem  $\left(\frac{2 + \sin x}{y + 1}\right) \frac{dy}{dx} = -\cos x$ ,  $y(0) = 1$ , then  $y\left(\frac{\pi}{2}\right)$  equals

(a)  $-\frac{1}{3}$

(b)  $\frac{1}{3}$

(c)  $-\frac{2}{3}$

(d)  $\frac{4}{3}$

(e)  $\frac{2}{3}$

## Answer Key

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| Q  | Ans | Brief explanation                                                                    |
|----|-----|--------------------------------------------------------------------------------------|
| 1  | (d) | Homogeneous linear system; the others miss $\mathbf{0}$ or fail a closure.           |
| 2  | (c) | $M_y = N_x$ forces $g'(x) = 0$ ; $g(0) = 1$ .                                        |
| 3  | (a) | $m^2 - 8m + 15 = 0$ : $m = 3, 5$ .                                                   |
| 4  | (a) | $T = 30 - 20e^{-kt}$ , $e^{-2k} = \frac{1}{2}$ , need $e^{-kt} = \frac{1}{4}$ .      |
| 5  | (c) | $\det = 4k = 0$ .                                                                    |
| 6  | (b) | $\rho = \sin x$ : $(y \sin x)' = \sin^2 x = \frac{1 - \cos 2x}{2}$ .                 |
| 7  | (d) | Linear in $x$ : $(xy)'_y = 2y \ln y + y$ , integral $y^2 \ln y$ .                    |
| 8  | (a) | $v = y/x$ : $xv' = (v + 1)^2$ , so $-\frac{1}{v+1} = \ln  x  - c$ .                  |
| 9  | (c) | $F = \frac{x^2}{y} + \frac{y^2}{x^3} + g(y)$ , $g' = y^{-1/2}$ .                     |
| 10 | (d) | $(xp)' = 4x$ , $p = 2x + \frac{A}{x}$ .                                              |
| 11 | (c) | $v = \sin^{-1} t$ ; by parts $x = t \sin^{-1} t + \sqrt{1 - t^2} + C$ , $C = 0$ .    |
| 12 | (d) | $\mathbf{u} \parallel (2, 1, 0, 0)$ , and $x_3 = 2x_4$ gives $l = 1$ .               |
| 13 | (b) | $u = y - 4x + 1$ : $u' = \sqrt{u} - 3$ .                                             |
| 14 | (c) | $v = 1/y$ : $(xv)' = -\ln x$ , $v = 1 - \ln x + \frac{C}{x}$ , $C = 1$ .             |
| 15 | (b) | $y + 1 = \frac{K}{2 + \sin x}$ , $K = 4$ , so $y(\frac{\pi}{2}) = \frac{4}{3} - 1$ . |